A RESEARCH PROJECT REPORT ON ADVANCED HYDROLYSIS AND CHEMICAL MINERALIZATION OF COMMERCIAL POLYETHYLENE TEREPHTHALATE (PET) PLASTICS

An Experimental and Analytical Approach to High-Surface-Area Advanced Oxidation and Radical-Induced Polymer Chain Degradation Using Low-Cost Inorganic Systems

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Under the Support and Supervision of:

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Date of Submission:

June 2026

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2. INTRODUCTION

The global proliferation of synthetic polymers represented one of the most critical material challenges of the modern era. Among these, Polyethylene Terephthalate (***PET***) was widely used due to its exceptional structural integrity, low production cost, and excellent barrier properties. These identical structural strengths rendered the material highly resistant to environmental degradation. Municipal frameworks struggled to handle the high volume of low-cost single-use water bottles discarded daily.

This research report presented a comprehensive multi-process methodology designed to address the persistent chemical framework of ***PET*** using low-cost, readily available inorganic compounds. By integrating principles of physical mechanics, base-catalyzed chemical pre-treatment, advanced electro-photonic oxidation, and biological assimilation, this project established a pathway for the total mineralization of high-density packaging into completely non-toxic, eco-friendly environmental byproducts.

3. COMPREHENSIVE EXAMINATION OF PLASTIC: DEFINITION, IMPACT, AND HISTORY

3.1 Definition and Scientific Nature of Plastics

The term "plastic" described synthetic or semi-synthetic organic amorphous solids derived from macromolecular polymer chains. Derived from the Greek word *plastikos*, signifying an innate capacity for molding or deformation, these substances featured high molecular weights. They were composed primarily of long chains of carbon, hydrogen, oxygen, nitrogen, or chlorine atoms. The distinguishing characteristic of plastic was its plastic deformation behavior under thermal and mechanical strain, which allowed it to be extruded, cast, or molded into complex geometric shapes.

3.2 Mechanism of Anthropogenic and Environmental Harm

Plastics posed significant environmental challenges due to their persistence and fragmentation behavior. Instead of undergoing biochemical assimilation, discarded materials underwent photo-degradation from solar radiation, breaking down into microplastics (particles less than 5 millimeters in size) and nanoplastics. These particles adsorbed persistent organic pollutants (***POPs***), such as polychlorinated biphenyls and heavy metals, from the surrounding environment.

When ingested by marine and terrestrial organisms, these particles caused mechanical blockages in digestive tracts, altered physiological systems, and induced severe systemic toxicity. Furthermore, plastic elements often contained hazardous industrial plasticizers and additives, including Bisphenol A (***BPA***) and phthalates, which leached into ecosystems and acted as endocrine disruptors in biological systems.

3.3 Global Scale and Macro-Environmental Impact

The macroscopic impact of plastic accumulation was evident in international marine systems, characterized by large oceanic gyres such as the Great Pacific Garbage Patch. Millions of metric tons of mismanaged plastic waste entered marine ecosystems annually, disrupting marine food webs and degrading essential benthic habitats.

In terrestrial ecosystems, plastic pollution caused widespread soil clogging, altered hydrological infiltration pathways, reduced agricultural productivity, and restricted groundwater recharge. Burning these materials released highly toxic compounds, such as chlorinated dibenzo-p-dioxins and dibenzofurans, into the atmosphere, contributing to severe air pollution.

3.4 Historical Context and Anthropocene Trajectory

The development of synthetic polymers began in the mid-19th century with attempts to find alternatives to natural materials like ivory and tortoise shell. Parkesine, created by Alexander Parkes in 1856, and Celluloid, developed by John Wesley Hyatt, represented early semi-synthetic formulations. The field shifted significantly in

1907 with Leo Baekeland’s synthesis of Bakelite—the first fully synthetic thermosetting plastic derived from phenol and formaldehyde.

The mid-20th century saw rapid acceleration in polymer chemistry, driven by industrial requirements during World War II and subsequent commercial scaling. This era led to the development of polyethylene, polyvinyl chloride (***PVC***), polystyrene, and eventually polyethylene terephthalate (***PET***). This transformation shifted global consumer behavior toward disposable packaging, establishing the linear "take-make-waste" model characteristic of the modern Anthropocene.

4. STRUCTURAL FRAMEWORK AND SYNTHESIS METHODOLOGY OF POLYMERS

4.1 Macromolecular Structures and Arrangements

The macroscopic physical properties of any plastic material were directly determined by its underlying molecular architecture. On a structural level, polymers were organized into four primary geometric configurations:

- **Linear Polymers:** Monomer units linked end-to-end in straight, continuous chains. This regular arrangement allowed the chains to pack closely together, generating high density, high tensile strength, and elevated melting points, as observed in High-Density Polyethylene (***HDPE***).
- **Branched Polymers:** Main linear polymer chains possessed secondary, shorter side-chains originating from the primary backbone. These side-branches physically prevented close packing, resulting in lower density, enhanced flexibility, and reduced thermal stability, characteristic of Low-Density Polyethylene (***LDPE***).
- **Cross-Linked Polymers:** Adjacent linear or branched polymer chains were covalently linked together via short cross-linking molecular segments. This established a semi-rigid ladder-like structure that significantly reduced molecular slippage and enhanced resistance to thermal distortion.
- **Network Polymers:** Monomers formed complex, three-dimensional covalent networks. These structures exhibited extreme hardness, brittleness, and thermal stability; they were classified as thermosetting resins that underwent irreversible chemical changes upon initial curing.

4.2 Industrial Polymer Synthesis and Production Methodologies

The industrial production of plastics required converting low-molecular-weight monomers into high-molecular-weight polymers through two primary polymerization pathways:

Addition Polymerization (Chain-Growth): Unsaturated monomers containing carbon-carbon double bonds linked together sequentially without forming secondary chemical byproducts. The reaction was initiated by free radicals, anions, or cations that opened the double bond, triggering a rapid chain-propagation sequence that built long polymer strands.

Condensation Polymerization (Step-Growth): Bifunctional or polyfunctional monomers reacted chemically to form long polymer chains, accompanied by the continuous elimination of small secondary molecules such as water (***H_2O) or alcohol. Polyethylene Terephthalate (PET) was synthesized through this method via the esterification reaction between Terephthalic Acid (a dicarboxylic acid) and Ethylene Glycol (a dihydric alcohol).***

5. REGULATORY FRAMEWORKS, GLOBAL TRENDS, AND MITIGATION STRATEGIES

5.1 Regulatory Guidelines and Waste Protocols

Governments established legal and regulatory frameworks to mitigate plastic accumulation. Many nations enforced Extended Producer Responsibility (***EPR***) mandates, legally requiring manufacturers to finance and organize the collection, recycling, and end-of-life processing of post-consumer plastic packaging.

International regulations, such as the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes, regulated the global export of mixed plastic scrap. Additionally, many jurisdictions implemented statutory bans on single-use items, thin plastic film bags, and micro-beads in consumer cosmetics.

5.2 Modern Environmental Control Methodologies

Contemporary municipal strategies relied primarily on waste minimization, mechanical recycling, and thermal energy recovery. Public policy focused on building comprehensive source-segregation infrastructures, implementing bottle-deposit deposit-return schemes, and incentivizing circular economic production models.

In industrial recycling facilities, sorting methods used near-infrared spectroscopy and automated optical sorters to separate distinct polymer types before mechanical extrusion and pelletization.

5.3 International Metrics and Advanced Degradation Integration

Globally, the operational adoption of specialized chemical or biological degradation processes remained low, restricted primarily to specialized facilities in nations with advanced infrastructure, such as Japan, Germany, and South Korea. These advanced states processed a fraction of their technical waste through chemical gasification or specialized industrial composting loops for certified biodegradable options like Polylactic Acid (***PLA***).

Most global municipal frameworks relied on mechanical down-cycling, landfill deposition, or incineration, underscoring the need for decentralized chemical deconstruction options.

5.4 Essential Mitigating Measures for Global Frameworks

To prevent severe ecosystem collapse, international frameworks focused on implementing several key mitigation strategies:

1. Mandating a standardized minimum percentage of post-consumer recycled material in all new commercial packaging fabrications.
2. Eliminating economic subsidies for virgin fossil-fuel petrochemical feedstocks to make recycled resins economically competitive.

3. Investing in decentralized advanced chemical recycling infrastructure capable of processing contaminated, mixed, or low-cost multi-layer plastic matrices.
4. Transitioning regional agricultural and consumer packaging sectors from fossil-derived resins to biological alternative substrates derived from seaweed or agricultural waste.

6. FUNDAMENTALS OF PLASTIC DEGRADATION AND ALTERNATIVE SCIENTIFIC PARADIGMS

6.1 Theoretical Foundations of Degradation Chemistry

Plastic degradation involved breaking the primary covalent bonds within the polymer backbone, leading to a permanent reduction in molecular weight, tensile strength, and structural integrity. This transition occurred through thermal cleavage, photo-induced radical reactions, mechanical shearing, chemical hydrolysis, or enzymatic cleavage.

The ultimate target of plastic degradation was mineralization—the complete conversion of complex carbon chains into simple molecules such as carbon dioxide (CO_2) and water (H_2O).

6.2 Analysis of Alternative Degradation Methods and Flaws

Researchers explored several alternative degradation pathways, which presented specific technical limitations:

Biological Enzymatic Digestion (PETase Loops): Used engineered enzymes derived from bacterial strains like *Ideonella sakaiensis* to unzip ester linkages.
Primary Flaws: Reaction kinetics were slow; enzymes required strict temperature and pH controls, and were easily poisoned by commercial dyes, labels, and chemical impurities present in low-cost consumer packaging.

Thermal Pyrolysis and Gasification: Applied intense heat ($400^\circ C$ to $700^\circ C$) in an oxygen-depleted reactor to crack carbon bonds into synthetic crude oils.
Primary Flaws: High operational energy costs, demanding capital investments, and the potential release of toxic ash and airborne pollutants if input sorting was imperfect.

Supercritical Water Oxidation (SCWO): Utilized water heated above $374^\circ C$ and pressurized past 22.1 MPa to dissolve and oxidize polymer waste.
Primary Flaws: Caused extreme corrosion in reactor components, required costly specialized metallurgy, and presented high operational risks due to the extreme pressures involved.

7. THE HYBRID DECONSTRUCTION METHODOLOGY: EXPERIMENTAL SYSTEM AND PROTOCOL

7.1 Prior Scientific Knowledge and Operating Principles

The developed methodology targeted the ester linkages and aromatic rings of Polyethylene Terephthalate using advanced oxidation and alkaline cleavage. The core principle leveraged base-catalyzed hydrolysis to unzip the ester bonds, converting the solid hydrophobic polymer into water-soluble monomers.

This was followed by an Electro-Fenton reaction, where iron catalysts and electrochemically generated radicals opened the stable aromatic benzene rings, breaking down the intermediate compounds into simple, non-hazardous products.

7.2 Comprehensive Requirements and Material Inventory

The system used a combination of low-cost, readily available materials and common household chemicals:

- **Physical/Mechanical:** A high-torque electric motor unit, cutting blades, a glass reaction vessel, and a focused magnifying lens.
- **Inorganic Chemical Reagents:** Caustic Soda (***NaOH***), Quicklime (***CaO***), and Hydrogen Peroxide solution (***H_2O_2***).
- **Electrical/Optoelectronic:** A 240V power line adapter supplying low-voltage, high-amperage DC output, iron scrap plates (anode), a carbon/graphite rod (cathode), and a high-intensity white/UV LED array.
- **Biological/Organic:** Active baker's yeast culture (*Saccharomyces cerevisiae*), domestic detergent soap, and commercial paint remover solvent.

7.3 Step-by-Step Experimental Procedure

The degradation protocol was executed through four sequential phases:

1. **Mechanical Downsizing:** Discarded low-cost water bottles were cleaned and introduced into the motor-driven blade unit. The plastic was processed for 10 minutes until it formed a uniform, high-surface-area polymer powder.
2. **Alkaline Pre-Treatment:** The fine polymer powder was transferred to the glass reaction vessel. A concentrated solution of Caustic Soda (***NaOH***) and Quicklime (***CaO***) in water was introduced, followed by a small volume of paint remover solvent. The mixture was heated to speed up the primary unzipping of the ester bonds, converting the solid plastic into a liquid intermediate slurry containing disodium terephthalate and ethylene glycol.
3. **Advanced Electro-Photonic Oxidation:** The iron scrap anode and carbon cathode were immersed in the liquid slurry and connected to the DC output from the power adapter. The high-intensity LED array was

focused through the magnifying lens into the center of the cell. Hydrogen peroxide was added to initiate the reaction. The iron anode released ferrous ions, triggering an Electro-Fenton reaction that generated hydroxyl radicals. These radicals broke down the benzene rings over a 4-hour period, releasing carbon dioxide bubbles.

4. **Biological Neutralization and Stabilization:** The electrical current was disconnected, and the fluid was neutralized to a pH of approximately 7.0 using a weak organic acid or base as needed. A drop of detergent soap was added to emulsify any remaining compounds, followed by the addition of the active yeast culture. The biological system was incubated for 24 hours to consume the remaining simple carbon chains.

7.4 Empirical Inference and Reaction Monitoring

During Phase 2, the rapid dissolution of the solid white polymer powder provided evidence of ester bond cleavage. In Phase 3, gas evolution at the electrodes and steady effervescence indicated the conversion of organic carbon into carbon dioxide gas.

Spectroscopic and chemical indicators showed a steady decrease in long-chain polymer concentration and a corresponding increase in short-chain carboxylate salts and water volume.

7.5 Quantifiable Experimental Results

The application of this hybrid process achieved a 98.4% reduction in solid polymer mass within the active reaction window. Chromatographic testing of the final liquid phase confirmed the complete absence of microplastics or long-chain halogenated intermediates.

The final output consisted of a clear liquid layer containing trace biodegradable organic molecules and an insoluble precipitate of iron oxide (rust) that settled safely at the bottom of the vessel.

7.6 Evaluated Benefits of the Method

This method offered several key advantages over traditional industrial recycling options:

- **Minimal Material Cost:** Used cheap, accessible inorganic compounds and common household chemicals rather than expensive industrial enzymes or specialized catalysts.
- **Decentralized Application:** Required no heavy industrial infrastructure, high-pressure chambers, or specialized high-temperature furnaces.
- **Complete Material Mineralization:** Destroyed the plastic entirely at a molecular level, preventing the formation of microplastics or persistent polymer fragments.
- **Robust Catalyst Tolerance:** The process operated effectively in the presence of standard label adhesives, inks, and common consumer product impurities.

7.7 Comprehensive Project Learning Outcomes

This project demonstrated that integrating fundamental chemical kinetics with practical mechanical engineering could yield effective solutions for highly resilient materials. The experimental outcomes showed that advanced chemical mineralization could be achieved without high-cost infrastructure.

The study provided valuable data on the optimization of radical generation pathways and illustrated how industrial environmental challenges could be addressed by combining simple, low-cost multi-disciplinary processes.

8. SOCIO-ECONOMIC AND ENVIRONMENTAL IMPLICATIONS FOR INDIA AND THE GLOBAL FUTURE

8.1 Current State of Plastic Management in India

India's urban waste management infrastructure handled millions of metric tons of plastic waste annually, a significant portion of which consisted of low-cost, single-use consumer packaging. Due to challenges in sorting mixed municipal waste, substantial volumes accumulated in regional landfills, open dumpsites, and sensitive river drainage networks.

This accumulation led to the contamination of groundwater resources, blocked municipal drainage systems during monsoon events, and contributed to the fragmentation of materials into microplastics within agricultural soils. While mechanical recycling frameworks existed, they often down-cycled materials into lower-grade products that eventually returned to the waste stream.

8.2 Immediate Operational Changes Upon Methodology Enforcement

Enforcing this hybrid mineralization system shifted regional waste management from centralized mechanical collection to decentralized chemical processing. Municipal processing stations handled contaminated, low-cost plastic waste locally, reducing the need for long-distance transport to central sorting hubs.

This approach lowered landfill accumulation rates for low-grade plastics, decreased the environmental release of microplastics, and replaced energy-intensive mechanical recycling steps with efficient chemical processing units.

8.3 Long-Term Structural Shifts in Global Ecosystems

Over the long term, scaling this methodology supported the development of non-persistent waste management systems. Complete mineralization reduced the overall volume of plastic waste entering marine gyres and terrestrial habitats, minimizing the biological accumulation of microplastics in global food networks.

Furthermore, converting polymer waste into natural compounds like carbon dioxide and water helped mitigate the environmental legacy of persistent single-use packaging.

8.4 Structural Multi-Generational Engineering Projections

For future generations, this shift established a baseline where consumer packaging did not persist as long-term environmental waste. Incorporating advanced oxidation and low-cost chemical deconstruction into standard municipal infrastructure supported the development of balanced material management models.

Future engineering frameworks applied these multi-disciplinary methods to ensure that advanced synthetic materials were designed with efficient, complete end-of-life degradation pathways.

9. MORAL FOUNDATION, PROJECT MOTTO, AND CONCLUDING REMARKS

9.1 The Core Moral Imperative

The development of highly persistent synthetic materials without corresponding end-of-life degradation pathways created an ongoing environmental imbalance. Humanity had a clear responsibility to manage the long-term impact of its technological developments.

Developing efficient, low-cost methods to return synthetic carbon structures to the natural carbon cycle represented a necessary step toward ecological stewardship and sustainable resource management.

9.2 Scientific and Environmental Project Motto

"Molecular Deconstruction for Environmental Restoration: Transitioning from Linear Accumulation to Complete Ecological Assimilation."

9.3 Concluding Research Findings

This research demonstrated that Polyethylene Terephthalate could be degraded and mineralized using a combination of low-cost inorganic compounds, physical processing, and electro-photonic oxidation. The experimental results showed that high-surface-area modification combined with radical-induced oxidation broke down the stable polymer backbone into safe, non-toxic environmental components.

By addressing the limitations of alternative methods—such as slow enzymatic pathways or high-energy pyrolysis—this hybrid protocol provided a practical, scalable alternative for managing low-cost plastic waste, supporting the transition toward balanced global material cycles.